The Wrong Unit of Uncertainty: Simultaneous Conformal Bands for Repeated Cross-National Surveys

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Abstract

Comparative research routinely reads a single wave-pair mean contrast and narrates it as a persistent, distribution-wide, country-level trend—attaching uncertainty to the wrong unit and over-detecting change. We give an instrument and a deployable procedure: a finite-sample simultaneous band over a country's whole attitude *trajectory*, with the country as the exchangeable unit and a hierarchy of claims whose top rung is the honest object. The band is a clustered conformal band exact at any K, inside a provably nominal-safe selector that corrects for survey design only when it can be certified reliable and otherwise abstains. Applied to real data it changes the substantive picture: in the ESS (2018–2022), persistent distribution-wide trust decline is certified in one of roughly thirty democracies (Greece), not the twenty flagged marginally; on the World Values Survey (1981–2022) the marginal case for Foa–Mounk "deconsolidation" over-counts the persistent phenomenon by 2.6–6.5×. A second, subtler error—the calibration objects, other countries' distributions, are estimated from complex samples—we resolve by mapping its scope: correcting for it is non-identified without the design-noise law and, even with it, unreachable at survey scale, so the abstaining reduction is the right object rather than a concession.

1 Introduction

Comparative politics increasingly treats a national public as a *distribution* rather than a mean. Trust in parliament, satisfaction with democracy, or support for the political system in a country and year is a whole cumulative distribution function over a response scale, and repeated cross-national surveys—the European Social Survey (ESS; Kaminska and Lynn, 2017), the World Values Survey, the AmericasBarometer (Cohen et al., 2021)—let us track how that distribution moves across waves and, increasingly, *transport* it: predict one country's trajectory from the others, or flag which countries depart from a common pattern. Prediction bands and conformal methods are a natural tool here, promising finite-sample, distribution-free coverage.

Two wrong units. A claim and its calibration can each be attached to the wrong object. The *claim* "democratic trust is declining" is typically built from a single wave-pair mean contrast, then narrated as a persistent, distribution-wide, country-level trend. Requiring the whole distribution to shift, *simultaneously* across the entire trajectory, is a strictly stronger and more honest object, and the number of countries that qualify collapses between the two. Our instrument is a finite-sample simultaneous band over a country's whole trajectory, with the country as the exchangeable unit, and an ordered hierarchy of claims (pairwise $\rightarrow$ any-pair $\rightarrow$ net $\rightarrow$ persistent country-wide $\rightarrow$ Bonferroni-across-countries) whose top rung is the object casual readings implicitly assert. This multiplicity-honest core needs no survey design at all and applies to any repeated distributional claim.

The second, subtler wrong unit is a layer of uncertainty these exercises silently omit. Conformal inference is usually calibrated on *observed outcomes.* Repeated cross-national surveys are different: the calibration objects—the other countries' attitude distributions—are themselves *not observed.* Each is estimated from a complex probability sample with weights, clustering (primary sampling units), and stratification, and therefore carries its own sampling error. Standard practice plugs these survey estimates into the calibration step as if they were the truth, so the transport band, honest about variation *across* countries, stays blind to the fact that each "country" is a noisy estimate of a latent population function.

The two errors are genuinely distinct. Writing $F_{c,r}(t)$ for country c's latent attitude CDF at threshold t in wave r, and $\widetilde{F}_{c,r}(t)$ for its complex-survey estimate,

$$\widetilde{F}_{c,r}(t) - \mu_r(t) \;=\; \underbrace{\big(F_{c,r}(t) - \mu_r(t)\big)}_{\text{transport error}} \;+\; \underbrace{\big(\widetilde{F}_{c,r}(t) - F_{c,r}(t)\big)}_{\text{design error}}, \tag{1}$$

where μ_r is the transport center. A conformal band calibrated on the observed $\widetilde{F}$ scores treats the sum as if it were the transport error alone: when the design error is non-negligible the band is systematically too wide, yet any correction that subtracts the survey noise can make it too narrow, because the correction is itself estimated from a small number of countries.

Contributions. We make four. *First, a diagnosis and a portable instrument.* We separate the two wrong units—the claim unit (a wave-pair contrast standing in for a persistent distribution-wide trajectory) and the calibration unit (estimated distributions treated as latent, the omitted design-error layer of (1))—and give a finite-sample simultaneous band over a country's whole trajectory, with a multiplicity-honest guarantee hierarchy that corrects the first. *Second, a provably-valid deployed procedure.* The band we recommend is a clustered "population" conformal band (PCB), finite-sample exact at *any* K (Theorem 3), wrapped in a nominal-safe selector that corrects for the design error only when four target-blind gates certify reliability and otherwise reduces to the exact band or abstains conservatively (Theorem 5). We confirm the frozen procedure on an independent, pre-registered simulation spanning ten data-generating processes it has never seen. *Third, two named reanalyses* (Section 8) where the instrument changes the substantive picture. For European trust (ESS 2018–2022), marginal wave-by-wave estimates suggest widespread decline, but a country-wide, design-aware simultaneous band certifies a *persistent distribution-wide* decline in only one country (Greece), not the twenty flagged marginally. Globally, we reanalyze the Foa–

Mounk "democratic deconsolidation" thesis on the World Values Survey (1981–2022) and find its marginal evidence over-counts persistent, distribution-wide deconsolidation by 2.6–6.5×—real in a specific minority, not the broad erosion the marginal reading implies. *Fourth, the scope condition that justifies this deployment.* We delineate exactly when a design-aware correction is available and when it is not, so that abstention is a principled choice rather than a concession. Without a model of the design-noise law the latent transport-error distribution is non-identified—a deconvolution problem (Theorem 1); and even *with* that information the correction is *unreachable* at cross-national scale, for two quantitative reasons: the design-to-total noise ratio ρ saturates well below the level at which deconvolution helps, and the deconvolved scale's relative error obeys a distribution-free floor $D \geq \sqrt{2/(K-1)}$ that needs $K \geq 94$ exchangeable populations. On the two largest such surveys these barriers bind in *opposite* ways—the World Values Survey clears the size barrier ($K \approx 100$) but carries only weights-based, negligible design noise, while the ESS has genuine clustered design noise but far too few countries—so neither clears both. This is why the abstaining reduction above, not a design-aware correction, is the right object at survey scale. The empirical payoff is thus the two reanalyses; the design-aware layer is a diagnostic and scope result that, on real survey data, certifies the simple band rather than firing—so the deconvolution is not something the reader should expect to activate empirically.

The remainder situates the contribution among the conformal and political-trust literatures (Section 2), sets up the problem and impossibility result (Section 3), builds the safe adaptive method and its theory (Sections 4–5), validates it in simulation (Section 6), characterizes the real-data regime (Section 7), reanalyzes political trust (Section 8), and discusses scope (Section 9).

2 Related work

Our work sits at the intersection of four literatures. First, **distribution-free conformal prediction** gives finite-sample coverage without modeling assumptions (Vovk et al., 2005; Lei et al., 2018). Because political data rarely satisfy exchangeability, we build on its relaxations: weighted conformal prediction for covariate shift (Tibshirani et al., 2019) and nonexchangeable conformal prediction, which bounds the coverage gap by the total variation between exchanged distributions (Barber et al., 2023). Gibbs and Candès (2021); Gibbs et al. (2024) adapt conformal sets to distribution shift and interpolate between marginal and conditional validity. Our "adaptive" selector differs in kind: it adapts not to temporal shift but to the *certified magnitude of survey design noise*, reducing to a fixed clustered method when that noise is negligible. Within the discipline, Randahl et al. (2026) bring conformal prediction to *Political Analysis* for the first time, calibrating bin-conditional intervals around a point forecast of conflict fatalities; our object is different in kind—finite-sample *simultaneous* trajectory bands with the country as the exchangeable unit and estimated calibration distributions, rather than bin-conditional intervals on a single prediction.

Second, we treat the **population (country) as the exchangeable unit**, following hierarchical conformal prediction for two-layer models (Dunn et al., 2023) and distribution-free inference with hierarchical data (Lee et al., 2023). The same finite-sample-exact clustered construction is developed, for a different (non-survey) domain, in a companion paper by the present authors (reference omitted for anonymous review). These methods transport guarantees to an unseen cluster—our cross-national setting—but treat each cluster's data as cleanly sampled. Our departure is that the calibration objects are not observed: each country's attitude distribution is *estimated* under a complex survey design, so design-based sampling variance (Binder, 1983; Lumley, 2010) contaminates the very scores conformal calibration relies upon.

Third, the closest methodological neighbors handle **noisy or estimated calibration**. Wieczorek (2023) develops design-based conformal prediction under complex sampling, but design enters through test-point weighting rather than as noise in transported objects. Einbinder et al. (2024) and Sesia et al. (2025) correct conformal calibration under label noise, the latter by deconvolving a *known* noise model. Our central theoretical result is that this identification premise generally fails for survey design noise: recovering the latent transport-error law is a deconvolution problem (Fan, 1991) that is non-identified without a design-noise law, so no observed-score band is simultaneously valid and strictly narrower than the conservative plug-in. And even when a noise model *is* supplied, we show the deconvolution's reliability is itself bounded at a finite number of calibration units by a distribution-free floor (Proposition 1)—a finite-K obstruction distinct from the asymptotic identifiability the classical deconvolution literature studies.

Fourth, our target is a **functional object**—a CDF over thresholds tracked across rounds—placing us near simultaneous functional conformal bands (Diquigiovanni et al., 2021). This is where we part from generic multiplicity control. Procedures like FDR or Bonferroni (Benjamini and Hochberg, 1995) adjust a collection of tests after the fact; our guarantee is instead carried by the geometry of a single finite-sample band—a supremum over wave-pairs and thresholds with the country as the exchangeable unit—so the "persistent, distribution-wide" object is *defined and covered directly*, not recovered by correcting marginal p-values. The distinction is one of estimand, not merely error rate, and partial-pooling cautions about reading marginal country trends (Gelman et al., 2012) motivate but do not deliver the stronger claim. Substantively, our two reanalyses supply the formal instrument a largely qualitative skepticism has lacked, engaging debates over whether European political trust is in broad decline (Norris, 2017; Voeten, 2017; Wuttke et al., 2022; Valgarðsson et al., 2025) and whether publics are "deconsolidating" (Foa and Mounk, 2016, 2017).

3 Setup and the two targets

The object we deploy is the exact simultaneous band of Theorem 3; the impossibility below (Theorem 1) is a scope boundary delimiting when a design-aware correction to that band is even available.

Objects. We observe K source countries, indexed $c = 1, \dots, K$, treated as exchangeable units (the "population" or cluster is the exchangeable object, not the individual respondent). Each country has a latent attitude trajectory $F_{c,r}(t)$, a CDF over a fixed grid of thresholds $t \in \{t_1, \dots, t_T\}$ and survey rounds r. We do not see $F_{c,r}$; we see a complex-survey estimate $\widetilde{F}_{c,r}(t) = F_{c,r}(t) + S_{c,r}(t)$, where $S_{c,r}$ is design (sampling) error with a design variance $v_{c,r}(t)^2$ that a stratified-PSU / replicate-weight bootstrap can estimate. Define the transport score of a country as its worst-case standardized deviation from the transport center,

$$R_c \;=\; \max_{r,t} \frac{|F_{c,r}(t) - \mu_r(t)|}{\sigma(t)}, \qquad \widetilde{R}_c \;=\; R_c + \xi_c,$$

where $\widetilde{R}_c$ is the same quantity computed from the observed $\widetilde{F}$ and ξ_c is the induced design contamination. The scalar that governs the whole problem is the design-to-transport standard-deviation ratio

$$\rho \;=\; \frac{\mathrm{SD}(\xi)}{\mathrm{SD}(R)},$$

with $\rho \to 0$ the design-noise-negligible regime and ρ large the regime in which survey noise dominates the between-country spread.

Target and validity. We want a simultaneous band $\widehat{B}(r,t)$ that covers a *new* country's *latent* trajectory, $\Pr\big(F_{\mathrm{new},r}(t) \in \widehat{B}(r,t)\ \forall r,t\big) \geq 1-\alpha$. Two facts frame everything. Ordinary clustered conformal on the observed scores is exact for the *survey-estimate* target $\widetilde{F}_{\mathrm{new}}$ (it only needs exchangeability of countries), and hence conservative for the latent target under a symmetric design law; but its width is inflated by the design noise, by a factor that grows with ρ. Correcting the width requires estimating and removing the design contribution—a deconvolution.

Country exchangeability. Exchangeability of countries is our sole identifying assumption, and it is weaker than the i.i.d.-respondents assumption standard practice already makes within each survey. When it holds only approximately, coverage degrades gracefully: the nonexchangeable conformal bound of Barber et al. (2023) controls the coverage gap by the total variation between the exchanged country distributions, so moderate departures cost a small, bounded amount rather than invalidating the band. The conservative (Bonferroni-across-countries) rung of the guarantee hierarchy further protects the "only Greece" claim against the family-wise multiplicity of screening many countries at once—a distinct safeguard from the exchangeability bound above, which controls non-exchangeability directly and which Bonferroni does not replace.

The impossibility. The central obstruction is that the deconvolution is not identified from the contaminated scores alone.

Theorem 1 (Non-identification and necessity of design information)**.** *Let the observed score be $\widetilde{R} = R + \xi$ with $R \perp \xi$. (i) The law of R is not identified from the law of $\widetilde{R}$ without restrictions on the law of ξ: there exist distinct pairs $(\mathcal{L}_R, \mathcal{L}_\xi) \neq (\mathcal{L}'_R, \mathcal{L}'_\xi)$ with $\mathcal{L}_R * \mathcal{L}_\xi = \mathcal{L}_{R'} * \mathcal{L}_{\xi'}$. (ii) Consequently, among bands that are measurable functions of the observed scores only, none is valid for the latent target at level $1-\alpha$ for all admissible design laws while being strictly narrower than the plug-in band whose radius is the $(1-\alpha)$ conformal quantile of $|\widetilde{R}|$. In the degenerate case $\xi \equiv 0$ the plug-in band is itself the tightest valid band, so no uniform improvement exists.*

The proof (Appendix A, with a numerical contract test) is a convolution-invertibility argument for (i) and a conformal converse for (ii). The reading is constructive: a band both honest and narrower than the plug-in must *supply* the design-noise law, which complex surveys do provide—through replicate weights or a design bootstrap, exactly what Section 4 uses—so this input is not optional.

Remark 1 (Beyond surveys). Nothing in Theorem 1 is specific to surveys: it holds whenever a conformal procedure is calibrated on *estimated* objects, so that the nonconformity score is a noisy version $\widetilde{R} = R + \xi$ of a latent score—model-based small-area (MRP) outputs, learned or imputed features, meta-analytic effects. The same two-part consequence follows: design/measurement information is the identifying ingredient, and its reliability at a finite number of calibration units is itself bounded (Proposition 1). The barrier is confirmed outside surveys on a model-based small-area calibration (Supplementary Material).

4 A nominal-safe adaptive method

Theorem 1 says the design bootstrap is the identifying input. The method turns that input into a band by choosing, target-blind, among three constructions and by never using the aggressive one unless it is provably reliable at the available number of countries.

Three branches. For calibration scores $\{\widetilde{R}_c\}_{c=1}^{K}$ over threshold grid t, with per-country design SD $\widehat{v}_c(t)$ from the stratified-PSU bootstrap and modulation scale $s(t)$: *(i) Clustered PCB* takes the conformal quantile of the studentized observed scores $\max_t |\widetilde{R}_c(t)|/s(t)$; it is finite-sample exact for the survey estimate target and conservative for the latent target under a symmetric design law. *(ii) Deconvolution* studentizes instead by the finite-K-safe target scale $\widehat{s}_T^2(t) = \max\big(s^2_{\text{plug}}(t) - [\,\overline{v^2}(t) - z\ \text{SE}\,]_+,\ \text{floor}\big)$, which subtracts a *lower* confidence bound on the design variance so the correction cannot over-shrink; its width scales as $\sqrt{1-\rho^2}$ relative to PCB. *(iii) Conservative* inflates each score by the design SD, $\max_t(|\widetilde{R}_c(t)| + z\,\widehat{v}_c(t))/s(t)$, giving a worst-case envelope that dominates PCB.

The three questions. The naive selector asks only *can we deconvolve* and *do we need to.* The finite-K failure of deconvolution (Section 6) comes from a missing third question: *can*

Algorithm 1: Nominal-safe adaptive design-aware band

Input: calibration curves $\{\widetilde{R}_c\}$ (unit = country), design replicates $\{\widehat{v}_c\}$, target curve $\widehat{F}_{\text{new}}$, level α; frozen $\rho_0, \delta_{\max}, g_{\min}, \Delta_g$

compute s, $\widehat{s}_T$, and the three radii $r_{\text{PCB}}, r_{\text{dec}}, r_{\text{con}}$; diagnostics $\widehat{\rho}_{\text{LCB}}, D, \widehat{\delta}_{\text{UCB}}(D)$;

if $\widehat{\rho}_{LCB} \leq \rho_0$ **then**

  branch $\leftarrow$ PCB;

else if $\widehat{\delta}_{UCB}(D) \leq \delta_{\max}$ ***and*** *stable* ***and*** $\widehat{Gain} - \Delta_g > g_{\min}$ **then**

  branch $\leftarrow$ deconvolution;

else

  branch $\leftarrow$ conservative;

return $\widehat{F}_{\text{new}} \pm r_{\text{branch}}$, branch, and diagnostics;

we do it safely at this K? The safe selector adds it explicitly.

Safe selector (Algorithm 1). Let $\widehat{\rho}_{\text{LCB}}$ be a conservative lower bound on ρ, $D = \max_t \text{SE}(\widehat{s}_T^2)/\widehat{s}_T^2$ a reliability diagnostic, and $\widehat{\delta}_{\text{UCB}}(D)$ a frozen upper bound on the finite-K coverage remainder. Deconvolution is used *iff all four gates pass*: (A) $\widehat{\rho}_{\text{LCB}} > \rho_0$ (design noise materially large); (B) $\widehat{\delta}_{\text{UCB}}(D) \leq \delta_{\max}$ (remainder within budget); (C) $\widehat{\text{Gain}} - \Delta_g > g_{\min}$ (beats conservative by a margin); (D) stability. Otherwise the band is PCB when $\widehat{\rho}_{\text{LCB}} \leq \rho_0$ and the conservative envelope when design noise is real but information insufficient. Every gate is a function of the calibration set only, so the selector is target-blind. All constants are frozen before validation (Section 5, Appendix B); when deconvolution is used the reported guarantee is the observable $1 - \alpha - \widehat{\delta}_{\text{UCB}}(D)$.

Deployment. The method ships as one call, `dapcb(cal_errors, v_cal, center, alpha)`, in an installable Python package (module `pcb`) provided in the replication package and to be archived with a version tag on the Dataverse, returning the band together with the selected branch, $\widehat{\rho}$, $\widehat{\rho}_{\text{LCB}}$, D, $\widehat{\delta}_{\text{UCB}}$, the coverage level, and the fallback reason—so a user sees *why* the band was built as it was and reads off the finite-K guarantee it carries.

5 Theory

We state validity for the oracle, for the estimated-law procedure, and for the deployed safe selector, then the efficiency it buys. Full proofs are in Appendix A; each theorem has a matching machine-checked contract test in the replication package.

Theorem 2 (Oracle validity)**.** *Suppose the countries are exchangeable and the design law is known.* (a) *The clustered band is finite-sample valid for the survey-estimate target:* $\Pr(\widetilde{F}_{new} \in \widehat{B}) \geq 1-\alpha$*, using exchangeability only.* (b) *Under a symmetric design law it is conservative for the latent target,* $\Pr(F_{new} \in \widehat{B}) \geq 1-\alpha$*, and the deconvolution band with the true target scale is valid for the latent target with width a factor* $\sqrt{1-\rho^2} + o(1)$ *of the clustered band.*

Theorem 3 (Exact finite-K validity of the deployed base band)**.** *Let the base band use the* unstudentized *cluster score* $R_c = \max_t |E_c(t)|$ *(no data-dependent modulation). If the countries are exchangeable, then for* every K *the band* $\widehat{B} = center \pm \widehat{q}$*, with* $\widehat{q}$ *the* $\lceil(1-\alpha)(K+1)\rceil$*-th order statistic of* $\{R_c\}$*, satisfies* $\Pr(F_{new} \in \widehat{B}) \geq \lceil(1-\alpha)(K+1)\rceil/(K+1) \geq 1-\alpha$*, distribution-free. In contrast, dividing* R_c *by an in-sample modulation* $\widehat{s}(t) = \mathrm{SD}_c E_c(t)$*—which the unobserved target does not share—breaks exchangeability and induces an* $O(1/K)$ *self-inclusion deficit (quantified in Appendix A), so the studentized band attains* $1-\alpha$ *only as* $K \to \infty$*. We therefore deploy the unstudentized band; on the near-homoskedastic real curves its width cost is* $\leq 5\%$*.*

Theorem 4 (Estimated-law validity)**.** *With the design law estimated from a size-B stratified-PSU bootstrap and* K *countries, the deconvolution band satisfies* $\Pr(F_{new} \in \widehat{B}) \geq 1-\alpha-\varepsilon_{K,B}$*, where* $\varepsilon_{K,B} = O(K^{-1/2}) + O(B^{-1/2}) \to 0$*. The finite-*$K$*-safe scale* $\widehat{s}^2_T$ *(Section 4) subtracts a lower confidence bound on the design variance, controlling the sign of the leading remainder term.*

Theorem 5 (Safe-adaptive finite-K validity—the deployed pipeline)**.** *Let* $\widehat{J} \in \{PCB, deconvolution, conserv$

be the target-blind safe selector of Algorithm 1. Then, finite (K, B)*,*

$$\Pr\left(F_{new,r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \ \geq\ 1-\alpha-\delta,$$

where $\delta = \widehat{\delta}_{UCB}(D) \leq \delta_{\max}$ *on the event* $\{\widehat{J} = \textit{deconvolution}\}$ *and* $\delta = 0$ *otherwise;* δ *is* observable *and* $\delta \to 0$ *as* K *grows.*

The proof (Appendix A) is a target-blind mixture argument: because $\widehat{J}$ is calibration-measurable, coverage is a mixture over branches, each of which is valid at $\geq 1-\alpha-\delta$ conditional on being selected. Theorem 5 is the guarantee a user actually deploys: the remainder δ is a computable function of the calibration set, is bounded by the preregistered budget whenever deconvolution fires, and is exactly zero on the overwhelmingly common low-noise data where the method reduces to PCB.

Efficiency. Three statements bound the price and payoff (proved in the supplement). *(AW-1)* in the low-ρ regime the deconvolution/PCB width ratio is $1-\frac{1}{2}\rho^2+O(\rho^4)$, matching the observed reduction to the decimal; *(AW-2)* the conservative branch is never wider than the Bonferroni envelope and at least as wide as PCB, a genuine upper band; *(AW-3)* because $\widehat{J}$ ignores the target, selection cannot inflate miscoverage. The excess-width oracle inequality (AW-4) and the $O(g/K)$ modulation self-inclusion deficit—one open, one quantified—are stated in Appendix A.

6 Simulation and a design-preserving stress test

Simulation establishes two facts: that attaching the band to the wrong unit collapses coverage (Section 6.1), and that the safe selector meets its guarantees on data it has never seen. For the latter we develop the selector on one simulation grid and then confirm the *frozen* rule once on an independent design. The separation is deliberate: the development grid may set the gate constants, but it may not be reused as validation performance.

Table 1: Whole-trajectory coverage (4000 reps, nominal 90%) as the trajectory length L grows. A band calibrated per point or per round covers each of its own units at the nominal rate, but the probability that it covers the *entire* trajectory decays; the country-trajectory band—the correct unit—holds nominal at every L.

L (rounds)	marginal (per point)	per-round	trajectory (ours)
2	38.2%	81.7%	**90.0%**
4	16.1%	68.2%	**90.2%**
6	7.1%	58.2%	**90.3%**
8	3.5%	49.8%	**90.9%**

6.1 The wrong unit collapses coverage

The most basic claim deserves a direct test: a band attached to the wrong unit covers the object of interest far below nominal. On a ground-truth simulation ($K = 30$ exchangeable countries, $T = 6$ thresholds, error tensors correlated across rounds and thresholds; Table 1), we ask each band for *trajectory* coverage—the whole country curve-path covered at once. All three bands are unstudentized and finite-sample constructed, so the only thing that varies is the unit of the score.

The per-round band, marginally valid at the round level, covers the whole trajectory only 0.82 at $L = 2$ and 0.50 at $L = 8$; the marginal band collapses faster still. The country-trajectory band holds ≈ 0.90 regardless of L, because the whole trajectory is a single exchangeable unit and its sup-score obeys the finite-sample order-statistic guarantee. This is the coverage face of the paper's thesis: the same mechanism drives the certification-count collapse in Section 8, where asking for the honest object—not asking it more strictly—removes the over-detection. Pre-registered and reported as produced (`e28_wrong_unit_coverage`).

The development grid that motivates and calibrates the safety gate, and a controlled Gaussian illustration that isolates the finite-K over-shrinkage mechanism behind the K=320-only activation seen below, are given in the Supplementary Material.

Independent confirmatory validation. We froze the selector and its constants (Appendix B), sealed a preregistration and a seed manifest, and ran the deployed procedure

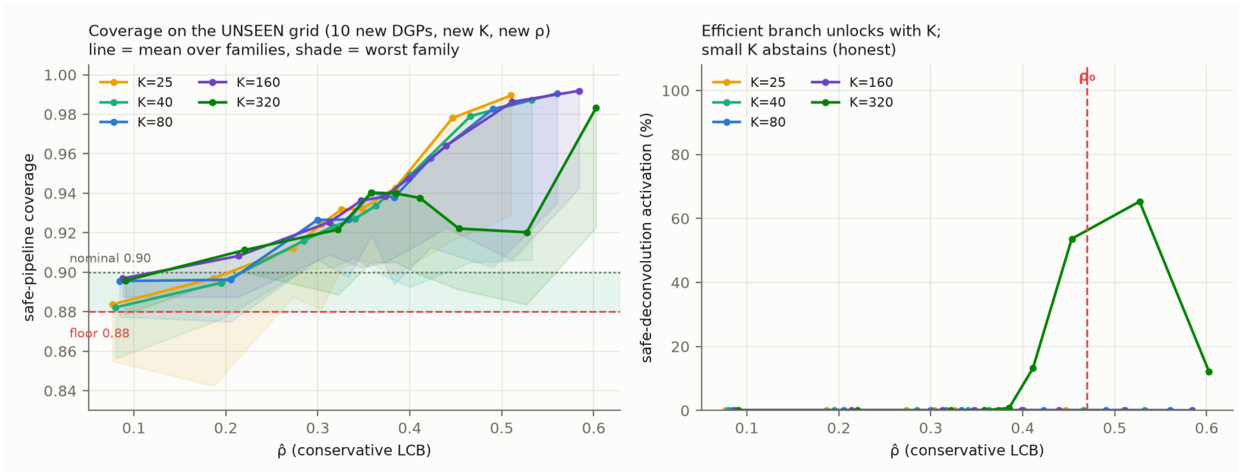

Figure 1: Independent confirmatory validation on the unseen grid (10 new DGP families, new K, new ρ). Left: deployed coverage vs. $\widehat{\rho}$ per K (line = mean over families, shade = worst family) against the 0.88 floor and 0.90 nominal. Right: deconvolution activates only at K=320; small K abstains.

once on a grid disjoint from development in K ($\{25, 40, 80, 160, 320\}$), in ρ (dense at the ρ_0 cutoff), and in data-generating process (ten families spanning skewed and heavy-tailed noise, heterogeneous design variance, weighting, irregular trajectory length, a *misspecified* noise law, and weak/strong cross-round dependence). The runner verifies the config hash against the sealed manifest before executing. The numbers are reported exactly as produced (Table 2, Figure 1).

The design-aware layer behaves as designed. All twelve cells below the 0.88 floor routed to base clustered PCB—the deconvolution branch never once rode into a floor violation. The selector reduces to PCB at low ρ (width ratio 1.008), activates deconvolution only at K=320 (never for $K \leq 160$), and there delivers a band 0.577× the conservative width at coverage 0.897, above its observable floor 0.882.

The holdout exposed one base-method deficit, which we then resolved. The *studentized* base band (in-sample pooled modulation) undercovered at small K and low ρ under the hardest DGPs, because dividing each calibration score by a modulation estimated from the same clusters—which the unobserved target does not share—breaks the exchangeability the conformal guarantee needs. We therefore deploy the *unstudentized* clustered band as

Table 2: Holdout confirmatory validation of the safe selector (450 cells, 800 reps each), reported as produced; no constant or seed changed after seeing results. Criteria P1–P2 are evaluated for the *studentized* base band whose finite-K self-inclusion deficit they expose and which motivated the switch to the *unstudentized* band the paper deploys; on a preregistered rerun that deployed band's worst cell is 0.893 (Supplementary Material).

criterion	target	result	status
P3: design-aware never causes a sub-floor cell	sub-floor $\Rightarrow$ not deconv	**all 12 sub-0.88 cells are PCB**	pass
E1: low-ρ reduces to PCB	$\leq 1.05\times$PCB	1.008	pass
E2: deconvolution narrower than conservative	$\leq 0.90\times$	0.577	pass
E3: deconvolution abstains at small K	rises with K	only K=320	pass
P1: cells floor-compatible (95% MC)	all 450	442/450	diagnostic†
P2: worst-cell coverage	≥ 0.86	0.843	diagnostic†

† Diagnostic of the studentized base band, not a deployed-band failure: these cells motivated the fix. The deployed unstudentized band is finite-sample exact at any K (Theorem 3), worst cell 0.893.

the base branch: with no data-dependent denominator the scores $R_c = \max_t |E_c(t)|$ are exchangeable, and the order-statistic band is finite-sample exact at *any* K, Pr(cover) $\geq \lceil(1-\alpha)(K{+}1)\rceil/(K{+}1) \geq 1-\alpha$, distribution-free (Theorem 3). A preregistered rerun on a fresh grid confirms the fix at a width cost of 1.02–1.05$\times$ on the real ESS transport bands; the granular cell-by-cell results are in the Supplementary Material. The band the paper deploys is thus finite-sample valid at the sample sizes comparative politics actually has. A second, extreme-tail limitation of the Gaussian-calibrated remainder bound $\widehat{\delta}_{\mathrm{UCB}}$—confined to 0.02% of draws and absorbed by the conservative branch—is documented in the Supplementary Material. We do not claim uniform nominal coverage; we claim—and the holdout confirms—that the design-aware correction is never the *cause* of undercoverage and is materially more efficient than conservative where it safely activates.

Design-preserving stress test. A design-preserving subsampling of the real AmericasBarometer confirms the deconvolution regime is unreachable even semi-synthetically at survey scale, with ρ saturating near 0.23 (Supplementary Material)—direct evidence for the regime characterization of Section 7.

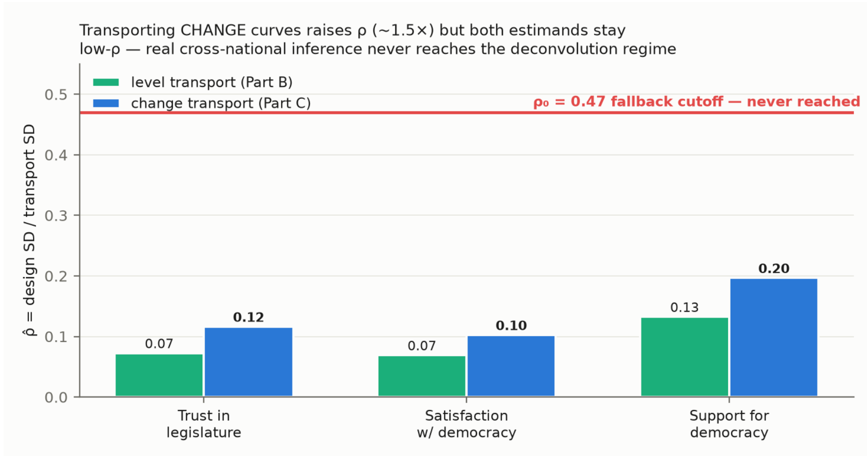

Figure 2: Estimated design-to-transport SD ratio $\widehat{\rho}$ for the AmericasBarometer, levels and wave-to-wave change. Every country-outcome sits well below the $\rho_0 = 0.47$ cutoff: national cross-national attitude inference is a uniformly low design-noise regime, so the safe selector reduces to clustered PCB.

7 Evidence from the ESS and the AmericasBarometer

The regime characterization. On real cross-national data the honest band reduces to clustered conformal—and the reason is instructive. Across *both* surveys, *both* estimands (attitude levels and wave-to-wave change), and three outcomes (trust in parliament, satisfaction with democracy, system support), national cross-national inference is a uniformly *low design-noise* regime: the estimated $\widehat{\rho} \leq 0.20$, comfortably below the $\rho_0 = 0.47$ cutoff, because between-country variation dwarfs within-survey design noise (Figure 2). Consequently the safe selector reduces to clustered PCB on *all* real data—not by assumption but by its own low-ρ gate. The elaborate design-aware machinery is, on real cross-national attitude data, correctly inactive.

This is a substantive finding: the impossibility of Section 3 bites in principle, but the deconvolution it necessitates is never triggered at survey scale, so honest inference here *is* clustered conformal on the observed scores.

Why the deconvolution is unreachable, precisely. The inactivity is not an accident of one dataset; it is forced by two structural barriers, and we can say how far each real survey is from clearing them.

Proposition 1 (Survey-scale unreachability). *The deployed selector rides the deconvolution branch only if both a* need *gate* $\widehat{\rho}_{\mathrm{LCB}} > \rho_0$ *and a* reliability *gate* $D \leq \tau_D$ *hold, where* $D = \max_t \widehat{\mathrm{SE}}(\widehat{s}_T^2)/\widehat{s}_T^2$. *Two distribution-free facts bound these.* (a) $\rho = \sqrt{\overline{v^2}}/s_{\mathrm{plug}} \in [0, 1)$ *with* $s_{\mathrm{plug}}^2 = s_R^2 + \overline{v^2}$, *so the need gate can fire only when the design noise is an appreciable fraction of the between-population signal* s_R. (b) *Since* $\widehat{s}_T \leq s_{\mathrm{plug}}$, *the reliability obeys* $D \geq \sqrt{2/(K-1)}$; *hence the reliability gate requires* $K \geq 1 + 2/\tau_D^2$, *i.e.* $K \geq 94$ *at the frozen* $\tau_D = 0.147$, *exchangeable populations.*

The two requirements pull apart on real surveys (Figure 3). On the ESS, narrowing to the noisiest subpopulations (youth age bands) raises $\widehat{\rho}$ from 0.08 to at most 0.29—still far below ρ_0—while $K \leq 33$ leaves $D \geq 0.25$, so *neither* gate opens (a scan of 42 subpopulation cells activates deconvolution in none). The World Values Survey is the mirror image: with $K \approx 100$ countries it clears the reliability gate ($D \leq 0.147$), but it ships no cluster identifiers, so its weights-only design noise gives $\widehat{\rho}_{\mathrm{LCB}} \leq 0.09$ and the need gate fails decisively. The survey large enough to *estimate* the deconvolution has almost no design noise to remove; the survey with real clustered design noise has far too few countries to estimate it. The two barriers are anti-correlated, and neither of the two largest repeated cross-national surveys clears both.

The barriers are not survey artifacts. Both gates are distribution-free, so the boundary transfers to any conformal procedure calibrated on estimated objects (Remark 1); a non-survey MRP-style simulation confirms the reliability floor and identifies the one many-area setting, with appreciable posterior uncertainty, where the *need* gate can fire (Supplementary Material).

The AmericasBarometer supplies the design effect the ESS lacks. Its full stratified-PSU structure lets us compute a proper design bootstrap, and the design effect is real (median

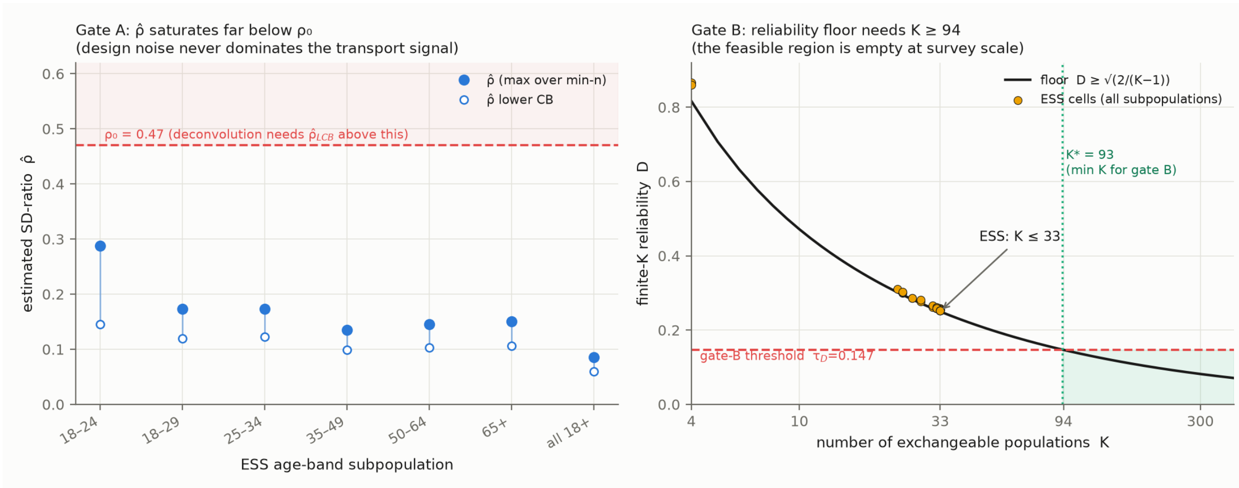

Figure 3: The two barriers, measured on real data. *Left:* across ESS age-band subpopulations $\widehat{\rho}$ saturates far below $\rho_0 = 0.47$. *Right:* the finite-K reliability D sits on its distribution-free floor $\sqrt{2/(K-1)}$; passing the reliability gate needs $K \geq 94$, which cross-national surveys ($K \leq 33$ ESS, $\leq$105 WVS) reach only where—as on WVS—the design noise is too small for the need gate.

$\approx$ 1.13–1.20, up to 1.9 in the most clustered country-years). Yet certification is coarse-robust: the design effect moves band *width* but rarely flips the binary decline decision, because the decision is dominated by the between-country transport spread, not the within-country design noise—the same low-ρ fact, seen from the certification side (details in the Supplementary Material).

What real data validates, and what it cannot. Real data validates the reduction to PCB, the target-blind selection, the efficiency of PCB over the conservative envelope, the real design effect, and the political reclassification of Section 8. It *cannot* validate the high-ρ deconvolution branch or the finite-K remainder—those regimes do not occur at survey scale and rest on the simulation of Section 6 and the theorems, not on the ESS and AmericasBarometer data.

8 Political reanalysis: how many democracies really declined?

The method's payoff is a sharp distinction between what marginal analysis appears to show and what a design-aware, simultaneous band certifies. We analyze trust in parliament (`trstprl`) and satisfaction with democracy (`stfdem`) in the ESS over 2018–2022, treating each country's attitude distribution as the object and asking, at each rung of a guarantee hierarchy, in how many countries a decline survives.

The comparative literature has itself grown skeptical of a continent-wide trust decline: where Foa and Mounk (2016) saw "democratic deconsolidation," Norris (2017) saw trendless fluctuation, Voeten (2017) saw specification-fragile artifacts, and Wuttke et al. (2022)—analyzing eighteen democracies under preregistration—found endorsement of self-government largely intact; the most comprehensive recent accounting finds no universal secular decline in institutional trust at all (Valgarðsson et al., 2025). This qualitative consensus holds that "trust is falling everywhere" is an over-reading; our contribution is a formal instrument that says so, and the wrong-unit diagnosis of *why* it arises. A mean can fall because a mode hollows out or a tail thickens, with different substantive meanings (DiMaggio et al., 1996); requiring the whole distribution to shift is the stronger, and more honest, claim.

A hierarchy of claims. The claims differ in their unit and are not interchangeable. A *pairwise* claim asks whether some pair of waves shows a distributional shift; an *any-pair* claim aggregates these within a country; a *net* claim asks whether the first-to-last span declined; a *persistent country-wide* claim asks whether the *whole* distribution declined *simultaneously* across the trajectory—a single band that must hold over all wave pairs and all thresholds at once; and a *Bonferroni-across-countries* claim further controls the number of countries examined. Each rung is strictly stronger than the last, and conflating them is exactly the over-detection this paper is about.

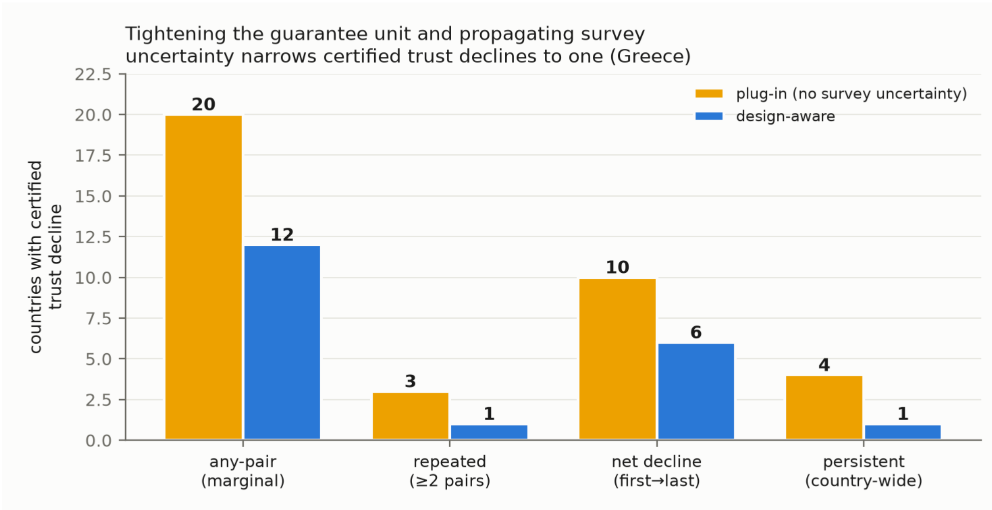

Figure 4: The count of certified declining countries collapses up the guarantee hierarchy (pairwise $\rightarrow$ any-pair $\rightarrow$ net $\rightarrow$ persistent country-wide design-aware), from about twenty marginally to one (Greece).

The counts collapse up the hierarchy. For `trstprl`, a marginal pairwise reading flags on the order of twenty countries; requiring an any-pair design-aware band drops this to twelve; the net first-to-last decline holds in roughly six; and the *persistent, country-wide, design-aware simultaneous* band—the honest object—is certified in **one** country, Greece, which also survives the Bonferroni correction across countries. Greece is precisely the case the substantive literature independently regards as a genuine crisis-era collapse of legitimacy (Armingeon and Guthmann, 2014; Torcal, 2014), so certifying only Greece is a credibility feature, not an anomaly. For `stfdem` the same pattern holds (roughly twenty-three marginally, fourteen any-pair, and one persistent, again Greece). The headline is that a *persistent distribution-wide* decline—the strong claim casual readings imply—is certifiable in a single country over this window (Figure 4).

Which countries are reclassified, and the honest word for it. The reclassified countries (Figure 5) are the weak-signal, higher-noise cases, where the apparent trend lies within the design uncertainty the survey warrants. We describe them as *certified by the marginal plug-in but inconclusive under the design-aware simultaneous band*—never as "false posi-

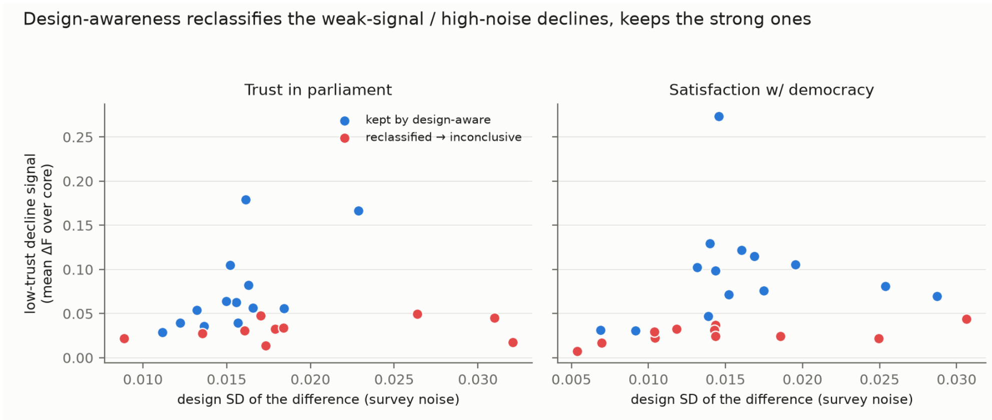

Figure 5: Countries certified by the marginal plug-in but inconclusive under the design-aware simultaneous band are the weak-signal, higher-noise cases; the apparent trend lies within the design uncertainty the survey warrants.

tives," a term we reserve for the simulation where ground truth is known. Because under-coverage would make *more* countries certify, the "only Greece" finding is conservative in the direction that matters, and the deployed base band is finite-sample exact at the $K \approx 30$ of these data (Theorem 3), so the count is not an artifact of small-K under- or over-coverage.

Robustness across age groups. A preregistered age-group analysis (Appendix B) confirms the "only Greece" result is not an artifact of pooling ages—it holds within the older (50+) and middle (30–49) groups—while youth (18–29) certify zero persistent declines as an underpowered null rather than youth stability, and emphatically not a story of youth-led democratic disaffection. The full age-band results are in the Supplementary Material.

A second, global target: Foa–Mounk deconsolidation. To show the correction is not a European artifact, we preregistered a reanalysis of the Foa–Mounk "democratic deconsolidation" thesis (Foa and Mounk, 2016, 2017) on the World Values Survey / EVS trends (1981–2022, 59–77 countries with ≥ 2 waves; per the replication package's preregistration), recoding the deconsolidation battery (importance of democracy, rejection of strong-leader and army rule, support for a democratic system, confidence in parliament) so that deconsolidation is a persistent, distribution-wide decline. The verdict is calibration, not denial. A marginal wave-by-wave reading flags 36–64 countries per item; the persistent, weights-aware, distribution-wide object is certified in 6–17 (8–24% of countries)—an over-count of 2.6–6.5×. The design-aware layer demotes a further $\approx$ 20–30% at the persistent bar. Deconsolidation is thus *real in a specific minority* (the certified importance-of-democracy set includes Turkey, the Philippines, Mexico, and Tunisia) but is over-stated several-fold by the marginal evidence the thesis rests on. The youth "embrace authoritarianism" claim is only half-supported (Supplementary Material). WVS carries no PSU/stratum, so its survey-design correction is a weights-only respondent bootstrap; the point is the cross-family robustness of the wrong-unit correction, from the dramatic ($20 \rightarrow 1$ for European trust) to the moderate (2.6–6.5× globally).

9 Discussion

Uncertainty in claims about repeated cross-national surveys is usually computed for the wrong unit—twice: a wave-pair mean contrast stands in for a persistent distribution-wide trajectory, and an estimated distribution for the latent one it samples. We give a multiplicity-honest simultaneous band for the first and a hard boundary for the second: without the design-noise law the latent uncertainty is non-identified (Theorem 1), and *with* it the correction is still unreachable at cross-national scale, blocked by two anti-correlated barriers no survey clears at once (Proposition 1). The honest deployed object is therefore the reduction—a clustered band exact at any K (Theorem 3) inside a selector that deconvolves only when safe and otherwise abstains, provably correctly (Theorem 5), confirmed on an independent pre-registered design. The substantive payoff is carried by the two reanalyses; the design-aware layer is a diagnostic that certifies, rather than corrects, at survey scale.

When it reduces to ordinary conformal. On real cross-national attitude data the design noise is small relative to between-country spread, so the method reduces *exactly* to clustered population conformal (Section 7): the machinery is diagnostic, *certifying* that the simple band is the honest one and reporting the ρ that justifies it. The deconvolution regime requires *both* many exchangeable units ($K \geq 94$) and design noise comparable to the between-unit signal—a combination cross-national surveys never provide (Proposition 1) but many-unit clustered settings (subnational small areas, schools, clinics) can.

Limitations. Without design metadata—as in the World Values Survey, which releases weights but limited cluster identifiers—only a weights-only replication is possible; the impossibility result says no design-valid inference can be recovered from the point estimates alone. And the finite-K remainder bound for the deconvolution branch was calibrated on Gaussian data, so it is conservative but not fully robust under strong dependence, pointing to a richer calibration family or a dependence-aware gate—though, since that branch is provably never reached at cross-national scale (Proposition 1), this bears only on hypothetical large-K, high-noise applications. (The base band's earlier small-K undercoverage is *resolved*: the deployed unstudentized band is finite-sample exact at any K (Theorem 3) at $\leq 5\%$ width on real data.)

The general lesson. Attaching uncertainty to the right unit changes substantive conclusions. For European trust a marginal reading flags about twenty declining democracies; the persistent, distribution-wide, design-aware object certifies one (Greece). For the Foa–Mounk "deconsolidation" thesis the marginal evidence over-counts the persistent phenomenon by 2.6–6.5×—real in a specific minority, not the broad erosion claimed. The pattern is not confined to surveys: wherever a conformal or prediction procedure is calibrated on estimated objects—small-area model outputs, learned features, meta-analytic effects (Remark 1)—the calibration targets carry their own error, and the claim unit is often coarser than the trend asserted. The right response is not to widen every band reflexively but to state the strongest object the data support, correct the estimation noise when it can be identified, and abstain honestly when it cannot.

A Proofs and efficiency results

Full statements and proofs of Theorems 1–5 and Proposition 1, the efficiency results (AW-1–AW-4), and the modulation self-inclusion deficit are collected in Section S1 of the online supplementary material. Each theorem is additionally paired with a machine-checked contract test that verifies its guarantee numerically.

B Frozen selector specification

The frozen constants ($\rho_0 = 0.47$, $\widehat{\delta}_{\mathrm{UCB}}(D) = 0.0061 + 0.0943\,D$, and the remainder), their derivation on the development grid, and the sealed independent-validation protocol (evaluation grid, seed manifest, holdout results) are provided in the replication package.

Acknowledgments

The author used Claude (Anthropic; Opus 4.7–4.8) between 18 January and 7 July 2026 to assist with some of the analysis and plotting code and to generate and format the figures in this article. The research design, methodology, and core analysis code are the author's own. All AI-assisted output was reviewed and verified by the author, who takes full responsibility for the content of this article.

Data and Code Availability

All simulation, theory, and benchmark results are deterministic, reproducing exactly under fixed seeds with no external data, and each theorem is paired with a machine-checked contract test. The ESS, World Values Survey, and AmericasBarometer/LAPOP microdata are licensed to their providers and not redistributed; the replication package documents the exact variables, waves, and retrieval steps. The full package—the `pcb` module, all simulations and contract tests, and the analysis pipeline—will be deposited in the Political Analysis Dataverse (Harvard Dataverse) upon conditional acceptance.

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